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Enhanced conductivity of sodium *versus* lithium salts measured by impedance spectroscopy. Sodium cobaltacarboranes as electrolytes of choice†

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The development of new types of ion conducting materials is one of the most important challenges in the field of energy. Lithium salt polymer electrolytes have been the most convenient, and thus the most widely used in the design of the new generation of batteries. However, in this work, we have observed that Na^+ ions provide a higher conductivity, or at least a comparable conductivity to that of Li^+ ions in the same basic material. This provides an excellent possibility to use Na^+ ions in the design of a new generation of batteries, instead of lithium, to enhance conductivity and ensure wide supply. Our results indicate that the dc-conductivity is larger when the anion is $[Co(C_2B_9H_{11})_2]^-$, $[COSANE]^-$, compared to tetraphenylborate, $[TPB]^-$. Our data also prove that the dc-conductivity behavior of Li^+ and Na^+ salts is opposite with the two anions. At 40 °C, the conductivity values change from 1.05×10^{-2} S cm^{-1} (Li[COSANE]) and 1.75×10^{-2} S cm^{-1} (Na[COSANE]) to 2.8×10^{-3} S cm^{-1} (Li[TPB]) and 1.5×10^{-3} S cm^{-1} (Na[TPB]). These findings indicate that metallacarboranes can be useful components of mixed matrix membranes (MMMs), providing excellent conductivity when the medium contains sufficient amounts of ionic components and a certain degree of humidity.

Introduction

Hybrid organic–inorganic composite polymer electrolyte membranes, also known as mixed matrix membranes (MMMs), have attracted much attention because it is envisaged that they can overcome the problems associated with purely organic membranes. For instance, Nafion, which is the most widely used membrane electrolyte in direct methanol fuel cells, DMFCs,[1] suffers from high methanol permeability, due to the solvated protons;[2,3] thus, it is important that the methanol permeability through the electrolyte is reduced, while maintaining its good proton conductivity and mechanical stability.[4–6] To remedy the methanol permeability, hybrid organic–inorganic composite polymer electrolyte membranes have been produced, *e.g.*, with inorganic fillers[7,8] or inorganic acid-doped composite membranes,[8,9] but these are not without problems; although the inorganic fillers do restrict the methanol permeability, they cause poor proton conductivity. It is therefore important that novel components are developed to produce hybrid organic–inorganic composite polymer electrolyte membranes. The electrolyte membranes can be produced in a variety of architectures, which require mechanically and thermally stable inorganic backbones with good dielectric and conductivity characteristics that will vary depending on the sought applications, *e.g.*, fuel cells, batteries, supercapacitors, catalysis, controlled delivery, ion-conductors, *etc.*[10–13]

It is generally accepted that the selected inorganic phase should have the following properties: hygroscopic stability, high specific surface area, surface acidity, excellent compatibility with the organic polymeric phase and good conductivity. In this sense, many hygroscopic inorganic materials have been investigated to improve the membrane water adsorption and self-humidifying ability to work at moderate and high temperatures. It is known that water entrapped in the metal oxide provides thermal stability and increases the conductivity of the MMMs. This can be explained because there is an enhancement of the hopping, compared to the vehicular mechanism, which is more significant in the case of hydrated membranes at low temperatures.[14,15] The incorporation of heteropolyacids (HPA), metal phosphates such as zirconium phosphate salts and hydrogen salts ($MHXO_4$) is advantageous because of the increase in the porosity in the resulting MMMs.[16–19] For example, in hydrogen salts the conductivity at moderate temperatures varies from 10^{-2} S cm^{-1} for RbH_2PO_4 (around 140 °C) to 10^{-4} to $RbHSO_4$

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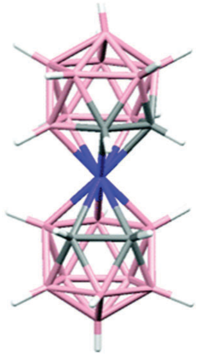

Fig. 1 Molecular representation of $[Co(C_2B_9H_{11})_2]^-$.

(around 160 °C).[20,21] Further, the characteristics of the central metal are very important for the conductivity and stability of the composites; for example, the surface acidity, the dimensions and the shape of inorganic particles notably influence the compatibility, the preparation of the composites and their permeability and conductivity performance.[15,22–26]

In this work, we focus on the moisture and temperature dependence of the conductivity of metallacarboranes. $[Co(C_2B_9H_{11})_2]^-$ represented in Fig. 1 is a sandwich compound that is anionic, with a very low charge density, reversibly redox electroactive, highly stable and with many possibilities for derivatization in a stepwise fashion. This has facilitated a wide range of E° values ranging from −1.80 to −0.35 V (*vs.* Fc), which are available today just by dehydrohalogenation.[27–29] Like ferrocene, $[Co(C_2B_9H_{11})_2]^-$ is an outer sphere electron transfer agent, but in contrast to Fc, its range of possible solvents is very wide, from low to high dielectric constants, thus broadening its possibilities for processability.[30] Despite being a purely inorganic molecule, it has a substitution behavior comparable or superior to organic frameworks.[31] This is why one would expect that the metallacarboranes would be adequate inorganic components for producing hybrid organic/$[Co(C_2B_9H_{11})_2]^-$ materials with innovative features.

Just as an organic polymer fragment has hydrophobic sites and has polar points, depending on the substituents, $[Co(C_2B_9H_{11})_2]^-$ has hydrophobic characteristics, but also polar behavior as has been demonstrated by its ability to produce vesicles, micelles and lamellae.[32–34] These properties, combined with the stepwise halogenations of the metallacarboranes that vary the degree of hydrogen and dihydrogen bonding to the organic polymer framework, effectively make these anionic molecules attractive for producing hybrid organic–inorganic composites for any desired application.

Organic–inorganic hybrid materials between $[Co(C_2B_9H_{11})_2]^-$ and polypyrrole or polythiophene have already been made. In the pyrrole polymerization process, a positively charged polymer is produced that is doped with the reversibly redox active $[Co(C_2B_9H_{11})_2]^-$. Indeed, the PPy/$[Co(C_2B_9H_{11})_2]^-$ represents an excellent example of the integration at the molecular level, being a genuine example of hybrid materials.[29,35–38]

These anions display strong non-bonding interactions with themselves and with polymers, inducing properties in addition to the individual characteristics of the components. Particularly relevant is the overoxidation resistance limit, ORL, which evidenced an improved resistance to the overoxidation, superior to the one offered by conventional anions.[37] These results and others led to the belief that these metallacarboranes can be useful components of MMMs and molecular electronics for a wide range of potential applications when they are successfully hydrated or immersed in humid environments. To this end, we have initiated studies to understand their ionic conductivity properties. In this paper, we report on the conductivity properties of the more common metallacarborane, $[Co(C_2B_9H_{11})_2]^-$, and compare the results with the conventional tetraphenylborate, $[B(C_6H_5)_4]^-$. Noticeably, both $[Co(C_2B_9H_{11})_2]^-$ and $[B(C_6H_5)_4]^-$ have the same charge and the same number of atoms, 45, but have a very different shape; the former is elongated, whereas the second is globular, and has binding capacity, and the former is able to generate hydrogen and dihydrogen bonding, but the latter is not. This is why the $[Co(C_2B_9H_{11})_2]^-$ can self-assemble, while $[B(C_6H_5)_4]^-$ cannot. We are of the opinion that a new class of Faradaic electrode materials, those of the metallacarboranes, may emerge as a molecular frontrunner in the field of energy storage systems. Similar to POMs, metallacarboranes are nanometric clusters with reversible redox activities, which can be used as building blocks for energy storage applications,[39] hydrogen fuel cells, catalysts for electrodes, electrolyte membranes, and hydrogen storage. discussion on challenges and perspectives in this field.[40–43]

Experimental

Materials

Lithium tetraphenylborate (Li[TPB]), H$[Co(C_2B_9H_{11})_2]$, (H[COSANE]), Na$[Co(C_2B_9H_{11})_2]$, (Na[COSANE]) and Li$[Co(C_2B_9H_{11})_2]$, (Li[COSANE]) were synthesized from commercial samples; Cs$[Co(C_2B_9H_{11})_2]$ and sodium tetraphenylborate (Na[TPB]) were obtained from Katchem Spol.sr.o and Sigma-Aldrich Co., respectively. Cationic exchange resin, strongly acidic PA, was obtained from Panreac.

Synthesis and preparation

The Na$[Co(C_2B_9H_{11})_2]$ was obtained, by means of cationic exchange resin, from Cs$[Co(C_2B_9H_{11})_2]$. Approximately 2/3 of the volume of the column (30 cm) was filled with the strongly acidic cationic exchange resin. Before starting, the cationic resin was kept at 24 h in 3 M HCl to hydrate it. Then, a 150 mL solution of HCl 3 M was slowly passed through the column to load it with H^+. To remove the excess HCl, distilled water was quickly passed through the column until neutral pH was reached. When the desired cation was sodium, a solution of 3 M NaCl was passed slowly through the column to exchange H^+ with Na^+ until neutral pH was reached (the change produced HCl). Distilled water was used to rinse the excess NaCl through the column. To know if NaCl was removed, 3 drops of a solution of 100 mM $AgNO_3$ was added to a small fraction of solution coming out of the column, until a clear solution was observed. Then, 30 mL of acetonitrile/water (50 : 50) mixture was allowed to flow through the column to set the column's liquid composition. Approximately 200 mg of the starting compound (Cs$[Co(C_2B_9H_{11})_2]$) was dissolved in a minimum volume of acetonitrile/water (50 : 50) and passed repeatedly (4 times) through the cationic resin. Before collecting the solution containing the metallacarborane, 50 mL of fresh acetonitrile/water (50 : 50) was added to the column. In a flask, 50 mL were collected, then the solvent was evaporated and the compound was dried in a vacuum.

The $Li[Co(C_2B_9H_{11})_2]$ was obtained by means of cationic exchange resin from $Cs[Co(C_2B_9H_{11})_2]$. The procedure was the same as for $Na[Co(C_2B_9H_{11})_2]$, but instead of loading the column with 3 M NaCl, 3 M LiCl was used. The lithium tetraphenylborate was obtained by means of cationic exchange resin from sodium tetraphenylborate. The same procedure as for $Na[Co(C_2B_9H_{11})_2]$ was used, but instead of loading the column with NaCl 3 M, 3 M LiCl was used.

The $H[Co(C_2B_9H_{11})_2]$ was obtained by liquid–liquid extraction from $Cs[Co(C_2B_9H_{11})_2]$. Approximately 200 mg of $Cs[Co(C_2B_9H_{11})_2]$ was dissolved in 20 mL diethyl ether. The sample was transferred to a separatory funnel and 15 mL of 1 M HCl was added. Two phases were formed and the metallacarborane sample passed through the organic layer with shaking. The organic layer was shaken three times with 1 M HCl (3× 15 mL) to completely change Cs^+ to H^+. The metallacarborane was collected in a flask and dried with anhydrous $MgSO_4$ to remove residual water. Then, the ether slurry was filtered, the diethyl ether was evaporated and the compound was dried in vacuum.

Characterization

The as prepared samples were analyzed by ATR-FTIR, ^1H NMR, $^1H\{^{11}B\}$ NMR, and MALDI-TOF.

Characterization of $H[Co(C_2B_9H_{11})_2]$, $Na[Co(C_2B_9H_{11})_2]$ and $Li[Co(C_2B_9H_{11})_2]$. ATR-FTIR (cm^{-1}): 3590, 3561, 3518 (brd, O–H), 3041, 3031 (wk, Cc–H), 2582, 2522 (shp, B–H). MALDI-TOF-MS: *m*/*z* calcd for $[Co(C_2B_9H_{11})_2]^-$: 324.76; found: 324.57. ^1H NMR (300 MHz, CD_3OCD_3): δ = 3.97 (br s, 4H, Cc–H). $^1H\{^{11}B\}$ NMR (300 MHz, CD_3OCD_3): δ 3.97 (br s, 4H, Cc–H), 3.41 (s, 2H, B–H), 3.02 (s, 2H, B–H), 2.73 (s, 4H, B–H), 1.97 (s, 4H, B–H), 1.59 (s, 6H, B–H). ^{11}B NMR (96 MHz, CD_3OCD_3): δ = 7.82 (d, 1J(B,H) = 144 Hz, 2B, B–H), 2.59 (d, 1J(B,H) = 135 Hz, 2B, B–H), −4.15 (m, 8B, B–H), −16.13 (d, 1J(B,H) = 155 Hz, 4B, B–H), −21.64 (d, 1J(B,H) = 167 Hz, 2B, B–H). $^{13}C\{^1H\}$ NMR (CD_3OCD_3): δ = 50.99 (s, Cc–H).

Sample preparation

We measured two types of samples: cobaltabisdicarbollide, H[COSANE], Li[COSANE], Na[COSANE], Li[TPB] and Na[TPB], each in dry and in wet conditions. Dry samples were prepared by pressing the raw powder into a thin disc, 12 mm diameter and about 0.7 mm in height. In order to prepare the wet samples, the raw material was first reground until a very fine powder was obtained. Then, 100 mg of this powder were mixed with 40 mg of ultrapure water, thereby obtaining a homogeneous wet paste, which retained its appearance throughout the measurement process in a Teflon ring, 10 mm in diameter.

Electrochemical impedance spectroscopy (EIS)

The conductivity of the samples in the transverse direction was measured in a range of temperatures between 20 and 120 °C by impedance spectroscopy in the frequency interval of $10^{-1} < f < 10^7$ Hz, applying 0.1 V signal amplitude. The measurements were carried out in dry and wet conditions. A Novocontrol broadband dielectric spectrometer (BDS) (Hundsangen, Germany) integrated by an SR 830 lock-in amplifier with an Alpha dielectric interface was used. The samples were previously dried and their thicknesses were measured afterwards using a micrometer, taking the average of ten measurements in different parts of the surface. For the study of the conductivity in dry conditions, the samples were dried in a vacuum cell. Next, the samples were sandwiched between two gold circular electrodes coupled to the impedance spectrometer acting as blocking electrodes. The assembly membrane-electrode was annealed in the Novocontrol setup under an inert dry nitrogen atmosphere before starting the actual measurement. To begin with, a temperature cycle from 20 °C to 120 °C to 20 °C, in steps of 10 °C, was done. Then, in a new cycle of temperature scanning, the dielectric spectra were collected in each step. This was performed to ensure the reproducibility of the measurements and to eliminate the potential interference of water retained by the composite membranes, taking into account the hygroscopicity of this kind of sample. For the measurements in wet conditions, the samples were previously dissolved in bi-distilled water in the ratio of 40 : 60 wt% (sample : water) to obtain a paste, which was subsequently sandwiched between the electrodes. During the measurements, the electrodes were kept in a BDS 1308 liquid device, coupled to the spectrometer and incorporating deionized water to ensure a fully hydrated state of the samples below 100 °C, and in equilibrium with the vapor above 100 °C, to simulate 100% RH atmosphere.[44]

During the measurements, the temperature was maintained for the isothermal experiments or changed stepwise by a nitrogen jet (QUATRO from Novocontrol) from 20 to 120 °C with a temperature error of 0.1 °C during every single sweep in frequency.

Results and discussion

Impedance spectroscopy measurements were carried out for all samples at different temperatures to obtain the conductivity. From the complex dielectric measurements, the conductivity was obtained by analyzing the data in terms of their dielectric permittivity $\varepsilon^* = \varepsilon' - j\varepsilon''$, where ε' and ε'' represent the real and imaginary parts of the permittivity and j the imaginary unity ($j^2 = -1$). The conductivity (σ_{dc}) can be calculated from the imaginary part when the Maxwell–Wagner–Sillars (MWS)[45–47] conditions are met, concisely when the bulk conductivity dominates as for a pure Ohmic conduction, with $\varepsilon'' = \sigma_{dc}/(\varepsilon_0\omega)$, where ε_0 represents the vacuum permittivity and ω the angular frequency of the applied electric field. This can be seen by the fact that the plot of $\log(\varepsilon'')$ *versus* $\log(\omega\ (s^{-1}))$ yields a straight-line with slope ≈ -1 over an extended range in the region of high frequency. This is the behavior of the samples reported here for all kinds of ions incorporated, and for all temperatures studied. From the intersection of this line with $\log(\omega) = 0$, we can calculate the dc-conductivity (σ_{dc}). Following this procedure, the conductivities of Li[COSANE], Na[COSANE], H[COSANE], Li[TPB] and Na[TPB] have been obtained in the interval from 20 to 120 °C.

Representative spectra of the analyzed samples are shown in Fig. 2. This figure shows the double logarithmic plot of the imaginary permittivity ε'' *versus* the frequency for the samples Li[COSANE], Na[COSANE], Li[TPB] and Na[TPB], at 50 °C, respectively. For all these samples, we have modeled the dielectric loss spectra as a function of the frequency with the piecewise function,[48] by which in the region of moderate frequencies the shoulder has been explained as a Debye relaxation, due to the macroscopic polarization of the ionic charges as a consequence of the electric field applied. This relaxation is characterized by a relaxation time τ_{EP} that depends on sample thickness, temperature and the type of polymer.[48–50]

$$\mathrm{Im}\varepsilon'' = \begin{cases} \dfrac{\sigma'}{\varepsilon_0\omega^n} + \dfrac{\Delta\varepsilon_{EP}\cdot\omega\cdot\tau_{EP}}{1+(\omega\cdot\tau_{EP})^2} & \text{if} \quad \omega \le \omega_c \\ \dfrac{\sigma_{dc}}{\varepsilon_0\omega} & \text{for} \quad \omega > \omega_c \end{cases} \tag{1}$$

where ω_c (= $2\pi f_c$) is a parameter that represents the cut-off frequency that presumably defines the onset of electrode polarization (EP). $\Delta\varepsilon_{EP}$ is the dielectric strength until the EP is fully developed, and the relaxation time τ_{EP} is defined[48] by the bulk resistance R_B and the interfacial capacitance C_{EP} by $\tau_{EP} = R_B{\cdot}C_{EP}$. The value of τ_{EP} is related to the mean time for an ion crossing from one electrode to another. We interpret ω_c as the frequency at which the macroscopic electrode polarization is practically irrelevant and therefore, the changes in the polyelectrolyte to a quasi-ideal conduction can be expressed by the second piece of the function described by eqn (1).[48,51] The conductivity values of the samples have been taken as the frequency where the imaginary part of the permittivity increases with decreasing frequency as $\varepsilon'' \sim f^{-1}$, exhibiting the typical contribution to the dielectric loss from electrical conduction due to the conductivity of the ionic charges. From this figure, we can see that the behavior of Li[COSANE], Na[COSANE], Li[TPB] and Na[TPB] samples at 50 °C is linear in the region of high frequencies, with a slope practically equal to −1 and a correlation coefficient of 0.999 for all samples and for all the temperatures studied. This is the typical contribution to the dielectric loss from electrical conduction, which means that the behavior of the samples for this interval of frequencies is that of a pure conductor.[48] For the sample H[COSANE] a similar behavior is observed and it is shown in the ESI† (Fig. S1).

From the intercepts of the straight line observed at the high frequency region, the value of the dc-conductivity for each temperature has been calculated.

In the region of mid-frequency, the behavior of ε'' passes through a maximum or a saddle point, then continues increasing with decreasing frequency. This shoulder of the complex permittivity shows a Debye-like shape, due to the electrode polarization (EP) effect, characterized by a relaxation time τ_{EP} that depends on temperature, sample width and the chemical structure of the materials and their properties. The maximum in ε'' represents the full development of the electrode polarization.[52–55] At low frequencies, the dependence is again linear, but with a slope less than unity. This is an indication of a dependence similar to what happens in the high frequency region, but with an expression $\varepsilon'' = \sigma'/(\varepsilon_0\omega^n)$,[51] in which n takes values between 0.5 to 0.7, but with a conductivity σ' associated with other types of ions such as impurities and some very slow ions, most probably the same cations (Li^+, Na^+, H^+), which are diffusing more slowly because the EP has almost completely built-up.[53]

Fig. 2 shows a comparison of the dielectric spectra for all samples at 50 °C. A close inspection of Fig. 3a clearly shows two very well differentiated regions. At the high frequency region, the real part of the conductivity shows a plateau that starts at frequency f_c, where the double logarithmic plot of the imaginary part of the permittivity *versus* frequency (Fig. 3c) exhibits a behavior of $\varepsilon'' \sim f^{-1}$ before reaching the maximum characteristic of the full development of electrode polarization (EP). At low frequencies, the dependence is again linear, but with a slope less than unity ($n < 1$), such as indicated above where the dependence of the complex permittivity shows a Debye-like shape, due to the electrode polarization effect.

Notice that at the frequency f_{EP}, where the phase angle reaches a value near 45° in the Bode plot of the real part of the conductivity (Fig. 3a), it corresponds to the maximum in the plot of $\log\varepsilon''$ *vs.* $\log f$ (Fig. 3c), and at this frequency the plot of $\log\sigma''$ *vs.* $\log f$ (Fig. 3b) reaches a maximum, indicative of the full development of the electrode polarization (EP) where the relaxation time τ_{EP} is determined by $\tau_{EP} = 1/(2\pi f_{EP})$. On the other hand, in Fig. 3d we can see that the real part of the dielectric permittivity, ε' at f_{EP} turns from the high frequency limit to the static value ε_s. It is noticeable that in the high frequency region (10^6–10^7 Hz), only the sample Li[TPB] reaches the plateau that corresponds to the values of $\varepsilon'(\infty) \cong 22$. This value is slightly larger than those reported as typical for many ionic liquids and glasses,[56–58] and also larger than the values found for samples of PPG-LiTFSI (11%).[49] Long distance charge transport across the interfaces produces a distributed MWS relaxation, reflected in the increase of the real part of the permittivity while decreasing the frequency in all the range of temperatures attributed to EP. This process may be credited to the increased deposition of charges at the interfaces of the polymeric materials. However, we have not observed the plateau for the other samples. In these systems, structural changes related to the polymer relaxations are controlled by the dynamics of the same ions that carry the charge, which implies a strong coupling of the ionic movement to the structural dynamics. From the results shown in Fig. 3, it is observed that all samples studied have dielectric spectra similar to the behavior of the low molecular weight ionic liquid monomers of 1-vinyl-3-pentyl imidazolium bis(trifluoromethylsulfonyl)imide (PVIM NTf_2).[56] On the other hand, the experimental observation of the dielectric response allows us to conclude for both $\dfrac{d\varepsilon''}{df} = 0$ and $\dfrac{d\sigma''}{df} = 0$ that the peaks observed in the variation of the imaginary part of the conductivity and permittivity appear at the same position in the frequency axis. The conductivity values obtained for all membranes and temperatures are given in Fig. 4.

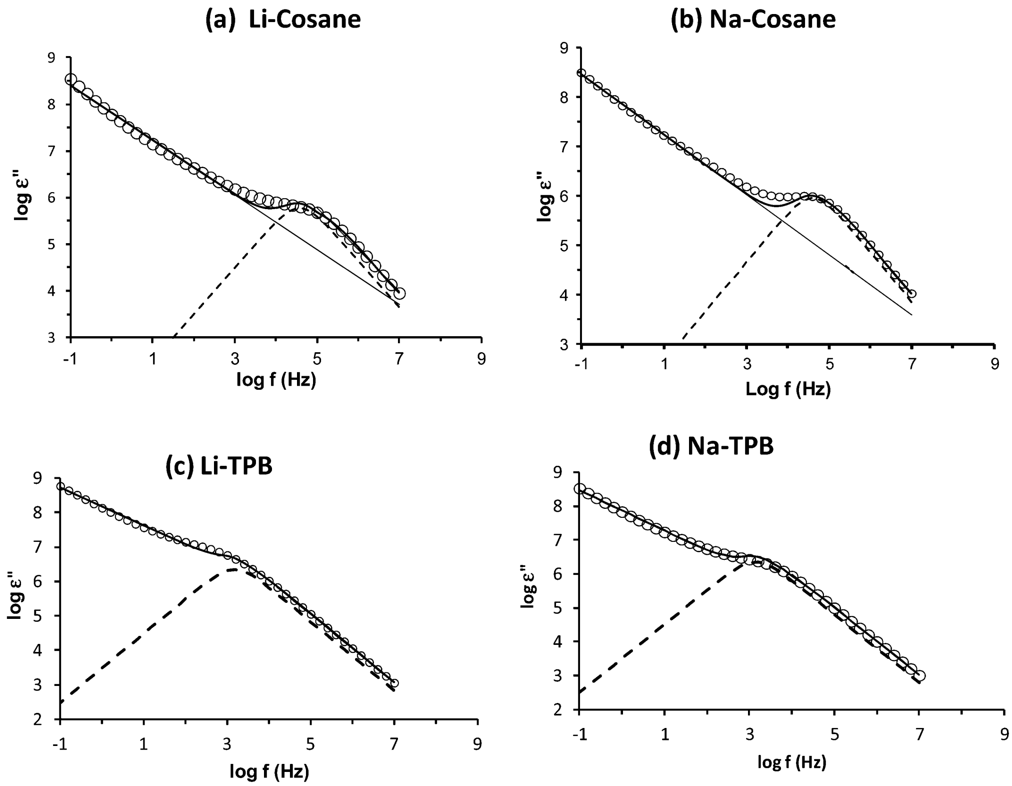

Fig. 2 Dielectric loss spectra as a function of the frequency, with the piecewise function for the samples: (a) Li[COSANE], (b) Na[COSANE], (c) Li[TPB] and (d) Na[TPB], at 50 °C. The experimental data for $\varepsilon''(f)$ are shown as open circles, the fitted electrode polarization contribution is plotted as a dotted line, and the solid line represents the fit of eqn (1). Also, we indicate the fit at low and high frequencies where we obtained the values of σ' and σ_{dc}, respectively, by means of a line (a and c). The exponent n for the fit in the low frequency region for all cases, is between 0.5–0.7 with a correlation coefficient of the adjusted r^2 = 0.99. The linear fit in the high frequency region has a slope between 0.98 and 1, with r^2 = 0.999, for the range of temperatures.

Fig. 4 shows the temperature dependence of the conductivity values. Most researchers in the field of polymer electrolytes assume that the temperature dependence of the conductivity follows an approximate Vogel–Fulcher–Tamman (VFT) equation expressed as follows:

$$\ln \sigma_{\mathrm{dc}} = \ln \sigma_{\infty} - \frac{B}{(T - T_0)} \tag{2}$$

where B is a fitting parameter related to the curvature of the plot that may be seen as the high temperature activation energy of the process underlying the dc-conductivity, σ_{∞} is the pre-factor related to the conductivity limit at higher temperatures and T_0 is the Vogel temperature considered as the one at which the relaxation time would diverge. In Table 1 we gathered the corresponding values obtained from the fitting curves shown in Fig. 4.

The mobile ion concentrations are either assumed constant with temperature or follow Arrhenius behavior. However, in most cases, the temperature dependence of mobile ion charges can be more complicated as occurs for $PEO^{-}Li^{+}$, $PEO^{-}Na^{+}$ and $PEO^{-}Cs^{+}$.[48] In our study, we can see that all samples can be described following a VFT equation, but in case of H[COSANE], Li[COSANE] and Na[COSANE], the linear tendency can be better described by two different slopes, above and below 70–80 °C, which differentiate two distinct regions. This is more distinguishable for H[COSANE] and less for Li[COSANE] and Na[COSANE], although a similar interpretation with two different slopes could be made for Li[COSANE] and Na[COSANE].

The conductivity of all samples is strongly humidity dependent, increasing by more than four or five orders of magnitude for all the samples studied. For example, for the sample Li[COSANE], the conductivity at 30 °C is around 10^{-7} S cm^{-1} (Fig. S2, ESI†), although at the same temperature in wet conditions, the conductivity is 3.1×10^{-2} S cm^{-1}. For all temperatures studied, the conductivity values increase with the temperature in the order σ(H[COSANE]) > σ(Na[COSANE]) > σ(Li[COSANE]).

The calculated activation energies from the VFT and Arrhenius equations follow the trends E_{ac}(H[COSANE]) = 5.6 kJ mol^{-1} < E_{ac}(Na[COSANE]) = 7.8 kJ mol^{-1} ≅ E_{ac}(Li[COSANE]) = 7.9 kJ mol^{-1}, considering all the ranges of temperatures. On the other hand, the activation energies for Li[TPB] and Na[TPB] are quite similar, with values of 2.3 and 3.5 kJ mol^{-1}, respectively. It is found that for Na[TPB] and Li[TPB], the conductivity decreases in the temperatures near 110–120 °C. For example, the conductivities at 110 °C and 120 °C are 3.5×10^{-2} and 1.2×10^{-3} S cm^{-1} for Li[TPB], whereas for Na[TPB] these values are 4.5×10^{-3} and 2.0×10^{-3} S cm^{-1}, respectively.

These E_{ac} values are lower than the corresponding ones obtained for the majority of low molecular weight ionic liquids (ILs)[56] and poly(ethylene oxide)-based sulfonated ionomers[48] with Li^{+} and Na^{+}. These results show that excellent ion mobilities

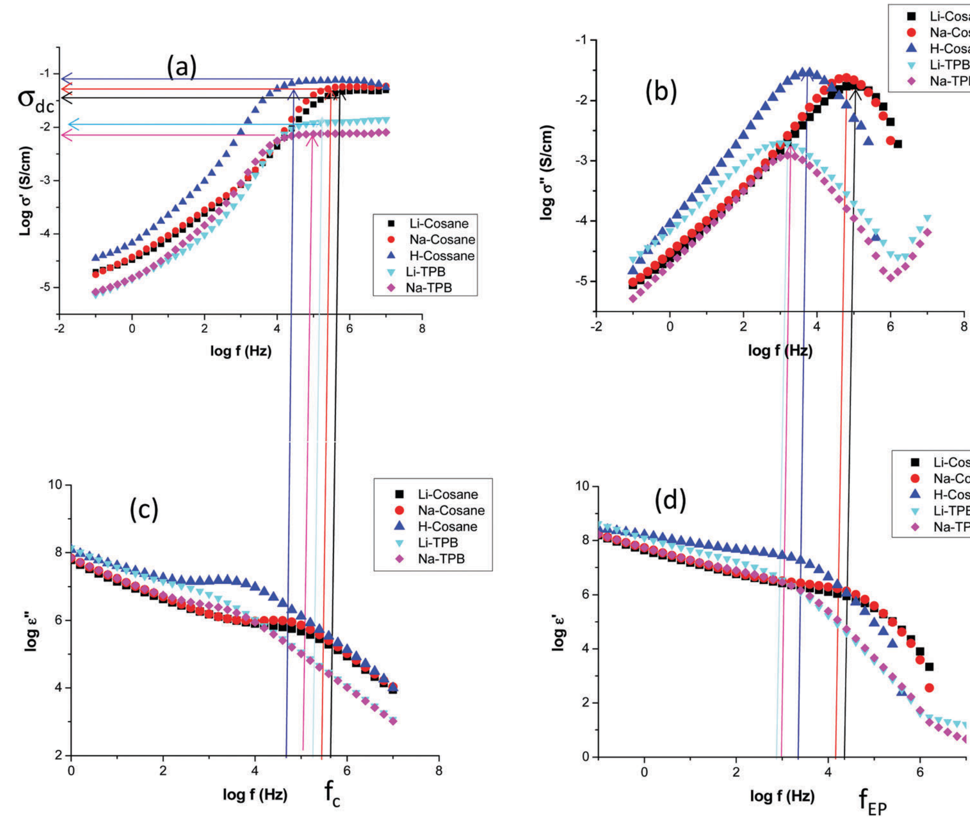

Fig. 3 Double logarithmic plot of real and imaginary parts of the complex dielectric function ε^*, and real and imaginary parts of the complex conductivity function σ^* *versus* frequency at 50 °C for Li[COSANE], Na[COSANE], H[COSANE], Li[TPB] and Na[TPB], respectively. The arrows in (a and c) represent the cut-off frequency (f_c) at which the conductivity has been observed in (a). This value is practically identical to that obtained from the plot of log(ε'') *versus* log(ω (s^{-1})) in the region of high frequency in (c). For the frequency (f_c), the plot of the imaginary part of the permittivity (c) exhibits a behavior of $\varepsilon'' \sim f^{-1}$. In (b and d) the arrows determine the frequency at which the full development of the electrode polarization is observed (f_{EP}). At this frequency (f_{EP}), we can see a maximum in (b) and a plateau tendency in (d), which is the result of the electrode polarization.

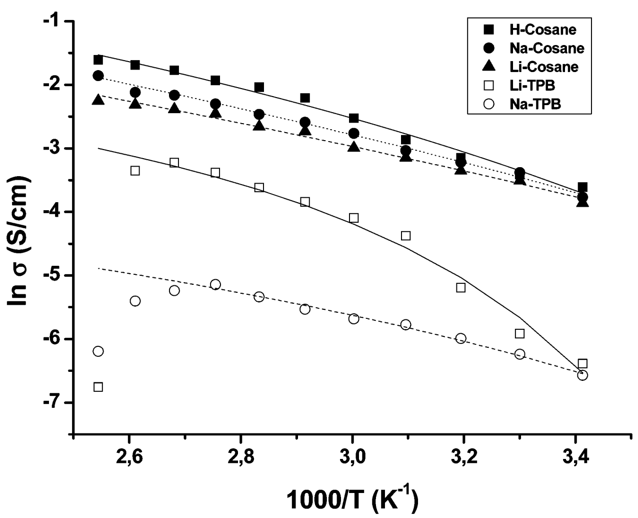

Fig. 4 Conductivity values obtained from the intercept of the plot of log(ε'') *versus* log(ω (s^{-1})) in the region of high frequencies when ω = 1. The solid line represents the fits of experimental data using VFT eqn (2) for [cation][COSANE] (cation = Li^+, Na^+ and H^+) and [cation][TPB] (cation = Li^+ and Na^+).

Table 1 Vogel–Fulcher–Tammann (VFT) parameters obtained from the fit of Fig. 4 for the samples analyzed in this study. The values of χ^2 parameters represent the sum of the square deviations between the experimental data and theoretical values obtained using eqn (2)

Sample	$\ln\sigma_\infty$ (S cm^{-1})	B (K)	T_0 (K)	χ^2	E_{act} (kJ mol^{-1})
H[COSANE]	1.38	680 ± 10	160 ± 2	0.06	5.6 ± 0.2
Li[COSANE]	0.93	950 ± 10	100 ± 2	0.03	7.9 ± 0.2
Na[COSANE]	1.4	940 ± 10	115 ± 2	0.01	7.8 ± 0.1
Li[TPB]	−1.2	280 ± 15	240 ± 2	0.02	2.3 ± 0.2
Na[TPB]	−2.7	425 ± 10	185 ± 3	0.01	3.5 ± 0.1

are obtained with these cobaltacarboranes in wet conditions. The incorporation of Li^+ instead of Na^+ produces a slight increase in activation energy, although both are almost identical for [COSANE]$^-$, and are slightly higher than those with [TPB]$^-$. The values of conductivity are similar to chemical structures containing carboxylic groups and inorganic metallic centers in their structures at ambient temperature and 100% RH, where the values found are in the range of 10^{-2}–10^{-3} S cm^{-1}.[59] However, our values are higher than the conductivity obtained in the nanocrystalline zeolitic imidazolate framework-8 (ZIF-8), which was around 4.5×10^{-4} S cm^{-1} at 94 °C and 98% RH.[60] The activation energy values for conduction are smaller in the metallacarborane samples than for ZIF-8, where the values found are surprisingly high, about 110 kJ mol^{-1}, more than thirty times these reported here (for $[COSANE]^-$ derivatives). Also, our results are lower (for the COSANE derivatives) than the values found for several ionic liquids [HMIM][BF_4] (molecular weight 254 g mol^{-1}), [HMIM]Br (molecular weight 247.18 g mol^{-1}), [HMIM]Cl (molecular weight 202.73 g mol^{-1}) and [HMIM]I (molecular weight 312.34 g mol^{-1}) and even smaller in the case of $[TPB]^-$ samples, where the values were around 2.9 kJ mol^{-1} and ten times lower than these obtained by Rivera and Rossler for different series of imidazolium based ILs.[61,62] As observed in Fig. 4, the dc-conductivity is larger when the monomer is $[COSANE]^-$ in place of $[TPB]^-$, although the behavior of Li^+ and Na^+ ions are opposed in each of the monomers. For example, at 40 °C the conductivity values are 1.05×10^{-2} S cm^{-1} for Li[COSANE], 1.75×10^{-2} S cm^{-1} for Na[COSANE], 2.8×10^{-3} S cm^{-1} for Li[TPB] and 1.5×10^{-3} S cm^{-1} for Na[TPB], *i.e.*, their change is about one order of magnitude, with the conductivity values being higher for Na^+ ion in the monomer $[COSANE]^-$, whereas the reverse happens for $[TPB]^-$. Both $[COSANE]^-$ and $[TPB]^-$ are monoanionic and have precisely the same number of atoms, 45. Taking into consideration the important changes in polarity and hygroscopicity associated with the change in the cation from Li^+ to Na^+ or H^+, the presence of retained water could have an influence on the conductivity observed. In this sense, the H^+ may be associated in the form of the hydronium ion, resulting in a greater mobility than Li^+ and Na^+ ions.

Fig. 5 shows a suggested mechanism for the conductivity of the salts of $[COSANE]^-$ and $[TPB]^-$, stressing the differences observed between the two monomers. Despite both $[COSANE]^-$ and $[TPB]^-$ being monoanionic and having precisely the same number of atoms, 45, and about the same number of H, 22 *vs.* 20, the conductivity of the H^+, Na^+ and Li^+ salts differ greatly in one or the second case. The main reason is the slightly hydridic character of H in $[Co(C_2B_9H_{11})_2]^-$, as compared to $[B(C_6H_5)_4]^-$. In $[COSANE]^-$, the negative charge is spread on the periphery of the molecule, whereas in $[TPB]^-$, it is more concentrated in the central boron atom. This permits that in $[Co(C_2B_9H_{11})_2]^-$, the $H^{\delta-}$ coordinates, even if it may be lightly, to the cations, whereas this does not occur with $[TPB]^-$. This is what is represented in Fig. 5, where in the metallacarborane case, the cation carries less load of coordinated water molecules because its coordination sphere is partially filled by the $H^{\delta-}$, whereas there is a high load of coordinated water in $[TPB]^-$. This leads to differences in the transport mechanism of conduction, tending more toward a hopping mechanism in the metallacarborane and a more vehicular case in the $[TPB]^-$ case.

In Fig. 6 the plot of electrode polarization time τ_{EP} as a function of the reciprocal of the temperature is shown. This curve cannot be described through a simple Arrhenius fit, indicating that in polymeric systems of low molecular weight a simple dependence on temperature is not generally present.

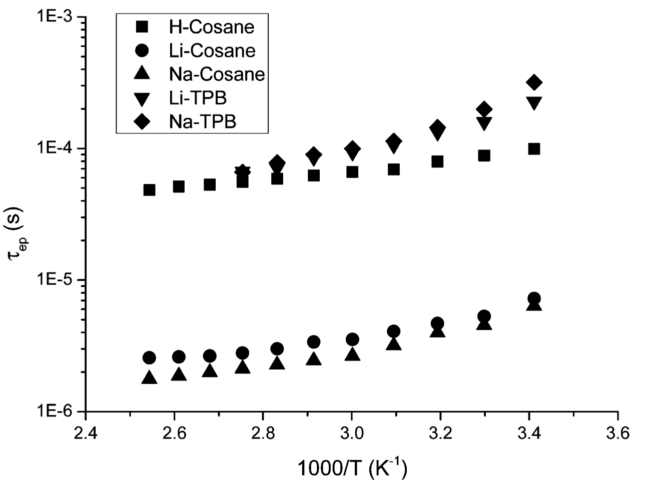

Fig. 6 Temperature dependence of the electrode polarization time τ_{EP}.

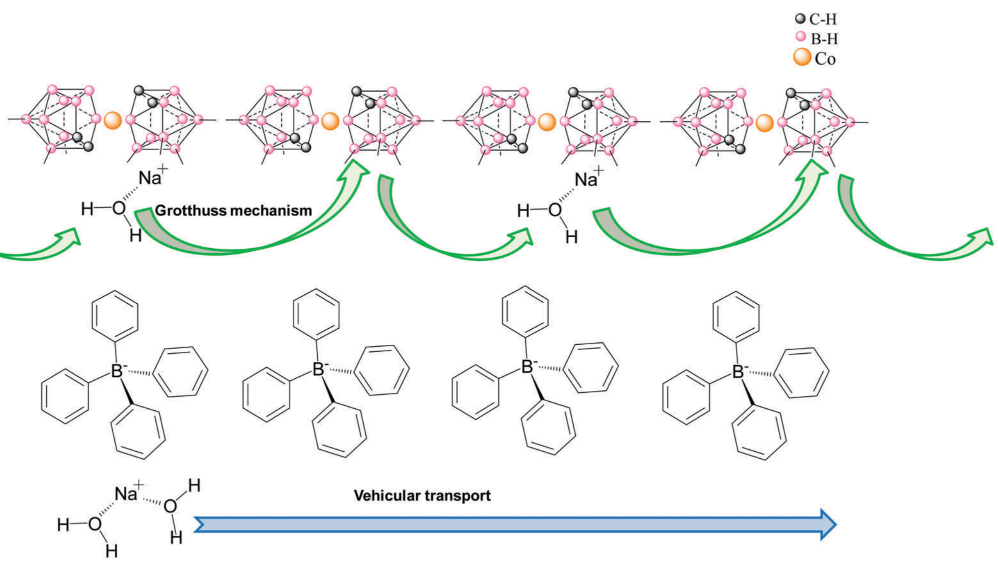

Fig. 5 Schematic representation of the proposed mechanism to explain the mobility of the cations through the monomers $[COSANE]^-$ and $[TPB]^-$. While in $[COSANE]^-$ the mechanism is through jumps (hopping mechanism), in $[TPB]^-$ the ionic transport is vehicular.

Taking into account that τ_{EP} represents the mean time for an ion to move from one electrode to another, at times longer than τ_{EP}, a large quantity of ions are deposited on the electrodes, generating an electric field that therefore retards the mobility of similar ions. In general, the polarization time of Li^+ and Na^+ ions is at least one order of magnitude smaller than protons in the case of monomer [COSANE]$^-$; however, this time it is larger for [TPB]$^-$ than [COSANE]$^-$ for both Li^+ and Na^+. The similarity of the behavior of Li[COSANE] and Na[COSANE] suggests that the change in counter cation, either Li^+ or Na^+, is not significant in all the ranges of temperatures, but this relaxation time increases when the cations are protons. This indicates that the counter cations affect the frequency at which the polarization starts to be significant in the system. The comparable behavior of the ions in [COSANE]$^-$ suggests that diffusivity is more effective when the ion is Na^+ and Li^+ than protons. However, a comparison between the monomers suggests that ion transport is more effective in [COSANE]$^-$ than [TPB]$^-$ for the same ions at the same temperature. It has been demonstrated that charge transport in aprotic ILs is basically characterized by two parameters, namely, the dc-conductivity σ_{dc} and some characteristic diffusion rate ω_c.[61,63,64] Likewise, we observe in our samples that structural relaxation is controlled by the dynamics of the same ions that transport charge. This is clear evidence that a strong coupling of ion motion to structural dynamics exists. However, in the equations corresponding to our samples, the characteristic diffusion rate is not calculated because the real part of the conductivity in the high frequency region finishes in a plateau.

Fig. 7 represents the activation plot for the cut-off frequency f_c for all samples studied. The temperature dependence of the characteristic rates of charge transport for the different cations and monomers shows that the activation diffusion rate has Arrhenius behavior.

The key parameter to describe the charge transport in eqn (1) is the cut-off frequency (f_c). This value represents the characteristic frequency at which the real part of the conductivity σ' begins to reach a plateau with decreasing frequency. The comparison between the samples for different types of ions permits the observation that the characteristic frequency of Li[COSANE] is slightly larger than Na[COSANE], but in both cases this frequency is about two times the value of H[COSANE]. When the monomer changes from [COSANE]$^-$ to [TPB]$^-$, the frequency at the electrode polarization begins to diminish considerably and is slightly lower for Li[TPB] than Na[TPB]. However at low temperatures, the characteristic frequencies for H[COSANE] and Na[TPB] are practically the same.

As we can see from Fig. 7, the temperature dependence of the characteristic rates of charge transport is approximately given by Arrhenius behavior

$$\ln f_c(T) = \ln f_\infty - \frac{E_{act}}{T}$$

where f_∞ is the cut-off frequency limit related to the relaxation rate at high temperature. E_{act} is the activation energy related to the relaxation rate of the process. The activation energies estimated from the Arrhenius behaviour follow the trend E_{ac}(H[COSANE]) = 5.30 ± 0.15 kJ mol^{-1} > E_{ac}(Na[COSANE]) = 4.63 ± 0.11 kJ mol^{-1} > E_{ac}(Li[COSANE]) = 3.73 ± 0.09 kJ mol^{-1}, considering all the intervals of temperatures. These values are comparable to those obtained by Krause *et al.*[61] for different imidazolium-based ionic liquids.

To learn about the correlation between conduction and dielectric relaxation, we proceeded to study the validity of the relation proposed by Barton–Nakajima–Namikawa (BNN),[64–67] given by the expression

$$\sigma_{dc} = B\varepsilon_0 \cdot \Delta\varepsilon \cdot \omega_c \quad (3)$$

where B is a constant of order unity, ε_0 is the vacuum permittivity, ε_s the static permittivity and ω_c is the cut-off angular frequency ($\omega_c = 2\pi f_c$). These scaling relations suggest that conduction and dielectric relaxation have their origins in one diffusion process. Fig. 8 shows the double logarithmic plot of the conductivity *versus* the angular frequency for all the samples studied.

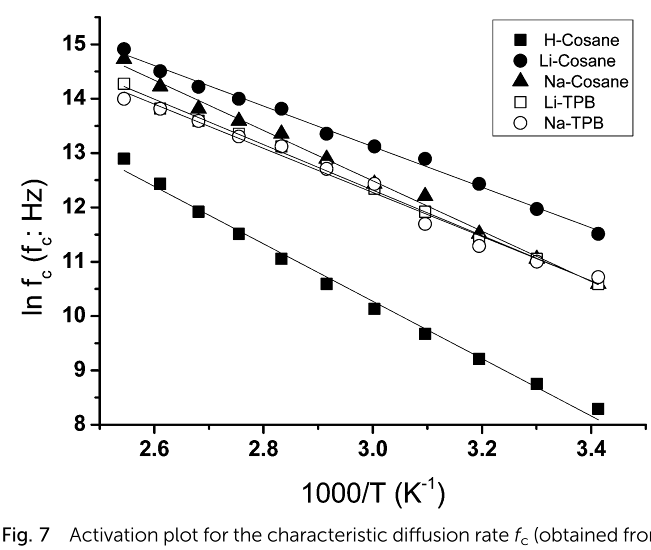

Fig. 7 Activation plot for the characteristic diffusion rate f_c (obtained from the plateau of the Bode plot of conductivity) for all samples.

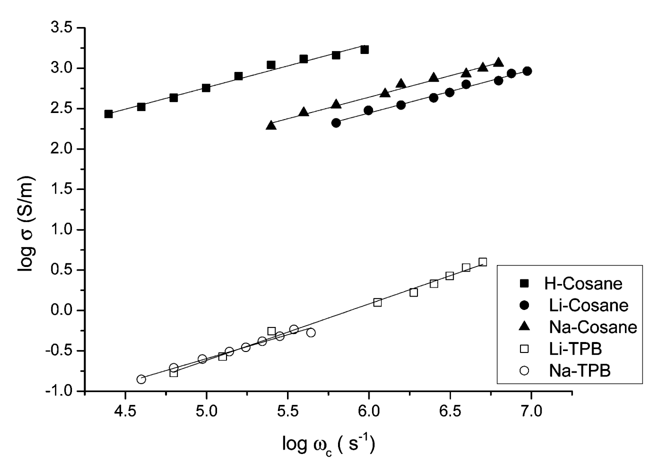

Fig. 8 Double logarithmic plot of the conductivity *versus* cut-off angular frequency. The solid line corresponds to the fit of the BNN relation (eqn (3)).

From Fig. 8 we can see the perfect scaling of the conductivity in accordance with eqn (3). From the fits of Fig. 8, we can estimate the dielectric strength, $\Delta\varepsilon$, considering that the value of the parameter B is equal to 1, (B is a non universal constant of order unity). The results obtained are 6×10^7, 1.5×10^7, 1.5×10^7 for the samples H[COSANE], Li[COSANE] and Na[COSANE], respectively, larger than the values for Li[TPB] and Na[TPB], which are near 5×10^6 in both samples. Further, if the static permittivity is known from experimental data, the constant B for each sample could be obtained. In our study, it has not been possible to determine the static permittivity from experimental values of dielectric spectroscopy impedance. This will be discussed in a future publication when the metallacarborane $[Co(C_2B_9H_{11})_2]^-$ and some of its halogenated derivatives, together with conventional tetraphenylborate $[B(C_6H_5)_4]^-$ will be part of an organic–inorganic framework as in a mixed matrix membrane (MMMs).

Conclusions

In this work we studied the temperature dependence of the ionic conductivity of different ions (Li^+, Na^+ and H^+) incorporated into the more common metallacarborane, $[Co(C_2B_9H_{11})_2]^-$, also known as $[COSANE]^-$, and compared the results with those of the conventional tetraphenylborate $[B(C_6H_5)_4]^-$, which has the same number of atoms.

The possibilities offered by the [cation]$[Co(C_2B_9H_{11})_2]$ (where [cation] = Li^+, Na^+, H^+) molecules are very attractive, particularly to produce hybrid organic–inorganic composite membranes of higher conductivity when water of hydration is present in its structure.

We have observed that in this metallacarborane, the conductivity of Na^+ ions is superior, or at least comparable, to that obtained for Li^+ ions. This allows the prediction of an excellent possibility for their use in the design of a new generation of batteries incorporating sodium ions. We have also found that the dc-conductivity is larger when the anion is $[COSANE]^-$, compared to $[TPB]^-$; however, the dc-conductivity behavior of Li^+ and Na^+ salts is opposite in the two anions. For example, at 40 °C the conductivity values are 1.05×10^{-2} S cm^{-1} for Li[COSANE], 1.75×10^{-2} S cm^{-1} for Na[COSANE], 2.8×10^{-3} S cm^{-1} for Li[TPB] and 1.5×10^{-3} S cm^{-1} for Na[TPB]. In general, the polarization time of Li^+ and Na^+ ions is at least one order of magnitude smaller than for the protons for $[COSANE]^-$; however, the polarization time is larger for $[TPB]^-$ than for $[COSANE]^-$, for both Li^+ and Na^+. The comparison between Li[COSANE] and Na[COSANE] suggests that with the change in counter cation, either Li^+ or Na^+ is not significant in all the range of temperatures, but the relaxation time increases when the cations are protons. This indicates that the counter cations affect the frequency at which the polarization starts to be present in the system. These results indicate that the diffusivity of ions in $[COSANE]^-$ will be higher than in $[TPB]^-$.

The slope of the ε'' *versus* frequency confirmed that ionic mobility at the higher frequency region is in accordance with the f_c observed from the Bode diagram for all the samples and temperatures studied. From the dielectric response, we can conclude that for both $\frac{d\varepsilon''}{df} = 0$ and $\frac{d\sigma''}{df} = 0$, the peaks or shoulder observed in the variation of the imaginary part of the conductivity and permittivity, respectively, appear at the same position in the frequency axe.

Finally it is important to note that metallacarboranes can be useful components of MMMs, providing excellent conductivity when the medium contains sufficient amounts of ionic component and a certain degree of humidity.

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References

1 E. Bakangura, L. Wu, L. Ge, Z. Yang and T. Xu, *Prog. Polym. Sci.*, 2016, **57**, 103–152.
2 F. Lufrano, V. Baglio, P. Staiti, V. Antonucci and A. S. Arico, *J. Power Sources*, 2013, **243**, 519–534.
3 S. P. Jiang, *J. Mater. Chem. A*, 2014, **2**, 7637–7647.
4 S. Meenakshi, S. D. Bhat, A. K. Sahu, P. Sridhar and S. Pitchumani, *Fuel Cells*, 2013, **13**, 851–861.
5 Y. Heo, H. Im and J. Kim, *J. Membr. Sci.*, 2013, **425**, 11–22.
6 A. K. Mishra, N. H. Kim, D. Jung and J. H. Lee, *J. Membr. Sci.*, 2014, **458**, 128–135.
7 G. Rambabu and S. D. Bhat, *Chem. Eng. J.*, 2014, **243**, 517–525.
8 S. Kango, S. Kalia, A. Celli, J. Njuguna, Y. Habibi and R. Kumar, *Prog. Polym. Sci.*, 2013, **38**, 1232–1261.
9 J. Zeng, B. He, K. Lamb, R. De Marco, P. K. Shen and S. P. Jiang, *Chem. Commun.*, 2013, **49**, 4655–4657.
10 S. J. Paddison, *Annu. Rev. Mater. Res.*, 2003, **33**, 289–319.
11 S. J. Peighambardoust, S. Rowshanzamir and M. Amjadi, *Int. J. Hydrogen Energy*, 2010, **35**, 9349–9384.
12 K. D. Kreuer, *Chem. Mater.*, 1996, **8**, 610–641.
13 K. D. Kreuer, S. J. Paddison, E. Spohr and M. Schuster, *Chem. Rev.*, 2004, **104**, 4637–4678.
14 C. Wang, E. Chalkova, C. D. Lute, M. V. Fedkin, S. Komarneni, T. C. M. Chung and S. N. Lvov, *J. Electrochem. Soc.*, 2010, **157**, B1634–B1642.
15 S. Hara, H. Sakamoto, M. Miyayama and T. Kudo, *Solid State Ionics*, 2002, **154–155**, 679–685.
16 Q. Yang, M. P. Kapoor and S. Inagaki, *J. Am. Chem. Soc.*, 2002, **124**, 9694–9695.
17 D. Margolese and J. Melero, *J. Chem. Mater.*, 2000, **12**, 2448–2459.
18 Y. G. Yin, S. Z. Qiao, Z. P. Xu, J. C. Diniz da Costa and G. Q. Lu, *J. Phys. Chem. C*, 2009, **113**, 3157–3163.
19 S. Fujita, A. Koiwai, M. Kawasumi and S. Inagaki, *Chem. Mater.*, 2013, **25**, 1584–1591.

20 V. G. Ponomareva, B. V. Merinov and V. V. Dolbinina, *Solid State Ionics*, 2001, **145**, 197–204.
21 M. Vijayakumer, A. D. Bain and G. R. Goward, *J. Phys. Chem. C*, 2009, **113**, 17950–17957.
22 S. Hara, S. Takano and M. Miyayama, *J. Phys. Chem. B*, 2004, **108**, 5634–5639.
23 Y. Kozawa, S. Suzuki, M. Miyayama and T. Okumiya, *Solid State Ionics*, 2010, **181**, 348–353.
24 R. R. Abbaraju, N. Dasgupta and A. V. Virkar, *J. Electrochem. Soc.*, 2008, **155**, B1307–B1313.
25 E. Chalkova, M. V. Fedkin, D. J. Wesolowski and S. N. Lvov, *J. Electrochem. Soc.*, 2005, **152**, A1742–A1747.
26 A. Sahu, G. Selvarani, S. Pitchumani, P. Sridhar and A. Shukla, *J. Electrochem. Soc.*, 2007, **154**, B123–B132.
27 P. González-Cardoso, A. Stoica, P. Farràs, A. Pepiol, C. Viñas and F. Teixidor, *Chem. – Eur. J.*, 2010, **16**, 6660–6665.
28 A. Pepiol, F. Teixidor, R. Sillanpää, M. Lupu and C. Viñas, *Angew. Chem., Int. Ed.*, 2011, **50**, 12491–12495.
29 M. Tarrés, V. S. Arderiu, A. Zaulet, C. Viñas, F. F. de Biani and F. Teixidor, *Dalton Trans.*, 2015, **44**, 11690–11695.
30 F. Teixidor, C. Viñas, A. Demonceau and R. Núñez, *Pure Appl. Chem.*, 2003, **75**, 1305–1313.
31 D. Olid, R. Nuñez, C. Viñas and F. Teixidor, *Chem. Soc. Rev.*, 2013, **42**, 3318–3336.
32 P. Bauduin, S. Prevost, P. Farràs, F. Teixidor, O. Diat and T. Zemb, *Angew. Chem., Int. Ed.*, 2011, **50**, 5298–5300.
33 D. Brusselle, P. Bauduin, L. Girard, A. Zaulet, C. Vinas, F. Teixidor, I. Ly and O. Diat, *Angew. Chem., Int. Ed.*, 2013, **52**, 12114–12118.
34 (*a*) P. M. Gassin, L. Girard, G. Martin-Gassin, D. Brusselle, A. Jonchere, O. Diat, C. Viñas, F. Teixidor and P. Bauduin, *Langmuir*, 2015, **31**, 2297–2303; (*b*) V. Dordovic, Z. Tosner, M. Uchman, A. Zhigunov, M. Reza, J. Ruokolainen, G. Pramanik, P. Cigler, K. Kalikova, M. Gradzielski and P. Matejicek, *Langmuir*, 2016, **32**, 6713–6722; (*c*) M. Uchman, V. Dordovic, Z. Tosner and P. Matejicek, *Angew. Chem., Int. Ed.*, 2015, **54**, 14113–14117.
35 (*a*) R. Nunez, I. Romero, F. Teixidor and C. Viñas, *Chem. Soc. Rev.*, 2016, **45**, 5147–5173; (*b*) R. Nunez, M. Tarres, A. Ferrer-Ugalde, F. F. de Biani and F. Teixidor, *Chem. Rev.*, 2016, **116**, 14307–14378.
36 (*a*) C. Masalles, S. Borrós, C. Viñas and F. Teixidor, *Adv. Mater.*, 2000, **12**, 1199–1202; (*b*) C. Masalles, S. Borrós, C. Viñas and F. Teixidor, *Adv. Mater.*, 2002, **14**, 449–452; (*c*) B. Fabre, J. C. Clark and M. G. H. Vicente, *Macromolecules*, 2006, **39**, 112–119; (*d*) E. Hao, B. Fabre, F. R. Fronczek and M. G. H. Vicente, *Chem. Mater.*, 2007, **19**, 6195–6205.
37 C. Masalles, F. Teixidor, S. Borrós and C. Viñas, *J. Organomet. Chem.*, 2002, **657**, 239–246.
38 C. Masalles, J. Llop, C. Viñas and F. Teixidor, *Adv. Mater.*, 2002, **14**, 826–829.
39 J. Suarez-Guevara, V. Ruiz and P. Gomez-Romero, *Phys. Chem. Chem. Phys.*, 2014, **16**, 20411–20414.
40 L. Carrette, K. A. Friedrich and U. Stimming, *ChemPhysChem*, 2000, **1**, 162–193.
41 C. Coma, H. García and F. X. L. I. Xamena, *Chem. Rev.*, 2010, **110**, 4606–4655.
42 H. J. Son, S. Y. Jin, S. Patwardhan, S. J. Wezenberg, N. C. Jeong, M. So, C. E. Wilner, A. A. Sarjeant, G. C. Schatz, R. Q. Snurr, O. K. Farha, G. P. Wiederrecht and J. T. Hupp, *J. Am. Chem. Soc.*, 2013, **135**, 862–869.
43 Y. Ren, G. H. Chia and Z. Gao, *Nano Today*, 2013, **8**, 577–597.
44 F. J. Fernández-Carretero, V. Compañ and E. Riande, *J. Power Sources*, 2007, **173**, 68–76.
45 J. C. Maxwell, *A treatise of Electricity & Magnetism*, Dover, NY, 1954.
46 K. W. Wagner, *Arch. Elektrotech.*, 1914, **2**, 371–387.
47 (*a*) K. W. Wagner, *Arch. Elektrotech.*, 1914, **3**, 67–106; (*b*) R. W. Sillars, *Proc. Inst. Electr. Eng.*, 1937, **80**, 378–394.
48 J. Klein, S. Zhang, S. Dou, B. H. Jones, R. Colby and J. Runt, *J. Chem. Phys.*, 2006, **124**, 144903.
49 R. Coelho, *Rev. Phys. Appl.*, 1983, **18**, 137–146.
50 J. R. MacDonald, *Phys. Rev.*, 1953, **92**, 4–17.
51 A. Munar, A. Andrio, R. Iserte and V. Compañ, *J. Non-Cryst. Solids*, 2011, **357**, 3064–3069.
52 Y. W. Che-Nan, F. Fan, J. R. Sangoro, M. B. Berman, S. G. Greenbaum, T. A. Zawodzinski and A. P. Sokolov, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2013, **87**, 04230l.
53 T. S. Sorensen, V. Compañ, R. Diaz-Calleja and E. Riande, *J. Appl. Phys.*, 1996, **79**, 403–411.
54 Y. W. Che-Nan, F. Fan, J. R. Sangoro, M. B. Berman, S. G. Grenbaum, T. A. Zawodzinski and A. P. Sokolov, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2013, **87**, 042308.
55 Y. Wang, F. Fan, A. L. Agapov, T. Sait, J. Yang, X. Yu, K. Hong, J. Mays and A. P. Sokolov, *Polymer*, 2014, **55**, 4067–4076.
56 J. R. Sangoro, C. Iacob, A. L. Agapov, Y. Wanf, S. Berdzinski, H. Rexhausen, V. Strehmet, C. Friedrich, A. P. Sokolov and F. Kremer, *Soft Matter*, 2014, **10**, 3536–3540.
57 H. Wiengärtner, *J. Mol. Liq.*, 2014, **192**, 185–190.
58 M. M. Huang, Y. Jiang, P. Sasisanker, G. W. Driver and H. H. Wiengärtne, *J. Chem. Eng. Data*, 2011, **56**, 1494–1499.
59 M. Sadakiyo, T. Yamada and H. Kitagawa, *J. Am. Chem. Soc.*, 2009, **131**, 9906–9907.
60 P. Barbosa, N. C. Rosero-Navarro, S. Fa-Nian and F. M. L. Figuereido, *Electrochim. Acta*, 2015, **153**, 19–27.
61 C. Krause, J. R. Sangoro, C. C. Iacob and F. Kremer, *J. Phys. Chem. B*, 2010, **114**, 382–386.
62 A. Rivera and E. A. Rossler, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2006, **73**, 212201.
63 J. R. Sangoro, M. Mierzwa, C. Iacob, M. Pluch and F. Kremer, *RSC Adv.*, 2012, **12**, 5047–5050.
64 J. R. Sangoro and F. Kremer, *Acc. Chem. Res.*, 2012, **45**, 525–532.
65 H. J. Namikawa, *J. Non-Cryst. Solids*, 1974, **14**, 88–100.
66 H. J. Namikawa, *J. Non-Cryst. Solids*, 1975, **18**, 173–195.
67 J. R. MacDonald, *J. Appl. Phys.*, 2010, **107**, 101101.